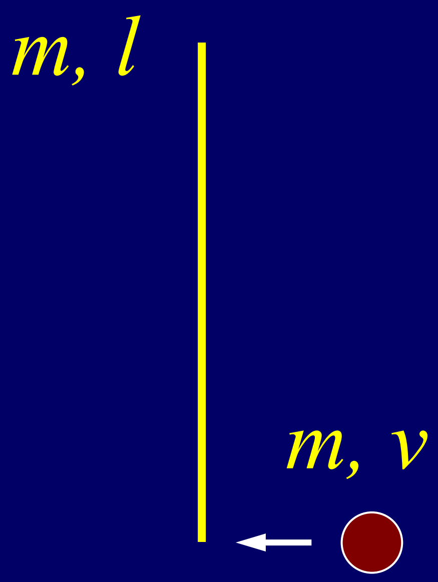
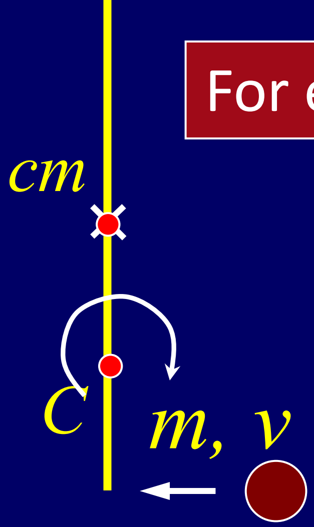
Quiz 8: A uniform rod of mass m and length l lies on a smooth horizontal plane. A ball of mass m moving at a speed v perpendicular to the length of the rod strikes it at the end of the rod. What are the velocity of the ball and the angular velocity of the rod after they collide if the collision is elastic.



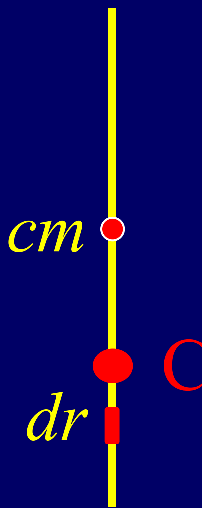


For elastic collision:

$$\frac{1}{2}mv^2 = \frac{1}{2}I_{cm}\omega^2 + \frac{1}{2}mv_{cm}{}^2 + \frac{1}{2}mv_f{}^2$$

$$\Sigma F_{ext} = 0, \quad \Delta\vec{p} = 0 \qquad mv = mv_{cm} + mv_f$$

$$\Sigma\tau_{ext} = 0, \quad \Delta\vec{L} = 0$$



about cm $$d\vec{p} = \lambda dr(\vec{v}_{cm} + \vec{\omega}\times\vec{r})$$

about C $$d\vec{p} = \lambda dr(\vec{v}_{cm} + \vec{\omega}\times\vec{r}_{cm,C} + \vec{\omega}\times\vec{r})$$

about cm $$\frac{mvl}{2} = \frac{mv_f l}{2} + \left|\int_{-l/2}^{l/2}\vec{r}\times d\vec{p}\right| = \frac{mv_f l}{2} + I_{cm}\omega$$

about C $$\frac{mvl}{4} = \frac{mv_f l}{4} + \left|\int_{-l/4}^{3l/4}\vec{r}\times d\vec{p}\right| = \frac{mv_f l}{4} - \frac{mv_{cm} l}{4} - \frac{1}{16}ml^2\omega + I_C\omega$$

about cm $\left|\int_{-l/2}^{l/2} \vec{r}\times(\omega\times\vec{r})\lambda dr\right| = I_{cm}\omega$ $I_{cm} = ml^2/12$

about C $\left|\int_{-l/4}^{3l/4} \vec{r}\times(\omega\times\vec{r})\lambda dr\right| = I_C\omega$ $I_C = 7ml^2/48$

about cm $$\frac{mvl}{2} = \frac{mv_f l}{2} + \left|\int_{-l/2}^{l/2} \vec{r}\times d\vec{p}\right| = \frac{mv_f l}{2} + \frac{1}{12}ml^2\omega$$

about C $$\frac{mvl}{4} = \frac{mv_f l}{4} + \left|\int_{-l/4}^{3l/4} \vec{r}\times d\vec{p}\right| = \frac{mv_f l}{4} - \frac{mv_{cm} l}{4} + \frac{1}{12}ml^2\omega$$

$$v = v_f + v_{cm}$$

Final solutions:

$v_f = 3v/5$ $v_{cm} = 2v/5$ $\omega = 12v/5l$